Code No: 136FF **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester (Special) Examinations, January/February - 2021
INDUSTRIAL MANAGEMENT
(Computer Science and Engineering)

Time: 2 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Discuss its role of planning in present day business organizations.
b) Critically examine Herzberg's two factor theory. [7+8]

2. What is committe organization? What are its benefits and limitations of organizational structures? [15]

3. Is layout design for a service organization any different from that of a manufacturing organization? Explain your answer in detail? [15]

4. Suppose an organization utilizes a variable based measurement system for process control. During a period, it was found that while all the plotted observations with in the control limits in the X bar chart, on point was lying outside the control limits in the R chart. What should the organization do in this case? [15]

5. Consider the following problem involving activities from A to J.

Activity	Immediate predecessor(s)	Duration (months)
A	-	1
B	A	4
C	A	2
D	A	2
E	D	3
F	D	3
G	E	2
H	F,G	1
I	C,H	3
J	B	2

a) Construct the CPM network.
b) Determine the critical path. [7+8]

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6. In a welding shop, a direct time study was done on a welding operation. One inexperienced industrial engineer and one experienced industrial engineer conducted the study simultaneously. They agreed precisely on cycle time (shown below) but their opinion on rating the worker differed. The experienced engineer rated the worker 100 percent and the other engineer rated the worker 120 percent. They used a 0.10 percent allowance fraction.

Cycle Time (Minutes)	Number of times observed
20	2
24	1
29	1
32	1

From the above statement,

a) Determine the standard time using the experienced industrial engineer's worker rating.

b) Find the standard time using the worker rating of inexperienced industrial engineer. [7+8]

7. Discuss the benefits, limitations of Job evaluation. [15]

8. Define departmentation? Explain the bases of departmentation. [15]

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